Steps to Work with BigQuery Data in Python on Mac/PC

Install Required Python Libraries

1. pip install google-cloud-bigquery pandas db-dtypes
2. brew install --cask google-cloud-sdk
3. initialize: gcloud init
4. Login with your Google account.

Authenticate Python with BigQuery

1. Run: gcloud auth application-default login
2. This creates credentials so Python can access BigQuery.

Import Libraries in Python

In [65]:
```
from google.cloud import bigquery
import pandas as pd
```

Create BigQuery Client and Convert Result to Pandas DataFrame

In [14]:
```
def get_table(table):
    return bigquery.Client(project="peronal-489013") \  # create a BigQuery client connected to project
    .list_rows(f"Athelete_Project.{table}") \           # read rows from dataset and table name passed in 'table'
    .to_dataframe()                                     # convert the BigQuery result into a pandas DataFrame
```

import the table from bigquery available in dataframe

In [40]:
```
df1 = get_table("athlete_events")
df2 = table("NOC_Regions")
```

Work with df Using Pandas or with other library

In [21]:
```
df1.head()
```

Out[21]:

	ID	Name	Sex	Age	Height	Weight	Team	NOC	Games	Year	Season	City	Sport	Event
0	107506	Hermann Schreiber	M	26	NA	NA	Switzerland	SUI	1936 Summer	1936	Summer	Berlin	Aeronautics	Aeronautics Mixed Aeronautics
1	23691	Heinz-Rudolf "Rudi" Cranz	M	17	NA	NA	Germany	GER	1936 Winter	1936	Winter	Garmisch-Partenkirchen	Alpine Skiing	Alpine Skiing Men's Combined
2	65820	Maurice Lafforgue	M	20	NA	NA	France	FRA	1936 Winter	1936	Winter	Garmisch-Partenkirchen	Alpine Skiing	Alpine Skiing Men's Combined
3	24115	Imre Csk	M	23	NA	NA	Hungary	HUN	1936 Winter	1936	Winter	Garmisch-Partenkirchen	Alpine Skiing	Alpine Skiing Men's Combined
4	117378	Lszl Szalay	M	21	NA	NA	Hungary	HUN	1936 Winter	1936	Winter	Garmisch-Partenkirchen	Alpine Skiing	Alpine Skiing Men's Combined

In [24]:
```
df2.head()
```

Out[24]:

	NOC	region	notes
0	AFG	Afghanistan	None
1	ALB	Albania	None
2	ALG	Algeria	None
3	ASA	American Samoa	None
4	AND	Andorra	None

Merge datasets to convert NOC codes into region names and prepare final dataset for analysis

In [41]:
```
df = df1.merge(df2, on="NOC", how="left")[
    ['ID','Name','Sex','Age','Height','Weight','Team','region','Games','Year','Season','City','Sport','Event','Meda
    ]
df.head()
```

Check the dtypes

In [42]:
```
df.dtypes
```

Out[42]:
```
ID          Int64
Name       object
Sex        object
Age        object
Height     object
Weight     object
Team       object
region     object
Games      object
Year        Int64
Season     object
City       object
Sport      object
Event      object
Medal      object
dtype: object
```

Convert the dtypes

In [43]:
```
df["Age"] = pd.to_numeric(df["Age"], errors='coerce').astype("Int64")
df["Height"] = pd.to_numeric(df["Height"], errors='coerce').astype("Int64")
df["Weight"] = pd.to_numeric(df["Weight"], errors='coerce').astype("float")
df.dtypes
```

Out[43]:
```
ID          Int64
Name       object
Sex        object
Age         Int64
Height      Int64
Weight    float64
Team       object
region     object
Games      object
Year        Int64
Season     object
City       object
Sport      object
Event      object
Medal      object
dtype: object
```

Lets Find and clean the data

In [44]:
```
df.head()
```

Out[44]:

	ID	Name	Sex	Age	Height	Weight	Team	region	Games	Year	Season	City	Sport	I
0	107506	Hermann Schreiber	M	26	<NA>	NaN	Switzerland	Switzerland	1936 Summer	1936	Summer	Berlin	Aeronautics	Aerona I Aerona
1	23691	Heinz-Rudolf "Rudi" Cranz	M	17	<NA>	NaN	Germany	Germany	1936 Winter	1936	Winter	Garmisch-Partenkirchen	Alpine Skiing	A S M Com
2	65820	Maurice Lafforgue	M	20	<NA>	NaN	France	France	1936 Winter	1936	Winter	Garmisch-Partenkirchen	Alpine Skiing	A S M Com
3	24115	Imre Csk	M	23	<NA>	NaN	Hungary	Hungary	1936 Winter	1936	Winter	Garmisch-Partenkirchen	Alpine Skiing	A S M Com
4	117378	Lszl Szalay	M	21	<NA>	NaN	Hungary	Hungary	1936 Winter	1936	Winter	Garmisch-Partenkirchen	Alpine Skiing	A S M Com

In [45]:
```
df[df.duplicated]
```

Out[45]:

	ID	Name	Sex	Age	Height	Weight	Team	region	Games	Year	Season	City	Sport
11223	44875	Alfrd (Arnold-) Hajs (Guttmann-)	M	50	<NA>	NaN	Hungary	Hungary	1928 Summer	1928	Summer	Amsterdam	Art Competitions
11229	49866	Emil Paul Anton Hoppe	M	52	<NA>	NaN	Austria	Austria	1928 Summer	1928	Summer	Amsterdam	Art Competitions
11230	44875	Alfrd (Arnold-) Hajs (Guttmann-)	M	50	<NA>	NaN	Hungary	Hungary	1928 Summer	1928	Summer	Amsterdam	Art Competitions
11231	61069	Charles Zeller Klauder	M	56	<NA>	NaN	United States	USA	1928 Summer	1928	Summer	Amsterdam	Art Competitions
11233	44875	Alfrd (Arnold-) Hajs (Guttmann-)	M	50	<NA>	NaN	Hungary	Hungary	1928 Summer	1928	Summer	Amsterdam	Art Competitions
...	...	...	...	...	...	...	...	...	...	...	...	...	...
194722	33564	William Edgar Exshaw	M	34	<NA>	NaN	Olle	UK	1900 Summer	1900	Summer	Paris	Sailing
194725	116681	Henri Lon Victor Susse	M	55	<NA>	NaN	Favorite-1	France	1900 Summer	1900	Summer	Paris	Sailing
194749	79412	Camille mile Gaston	M	32	<NA>	NaN	Turquoise-3	France	1900 Summer	1900	Summer	Paris	Sailing

	ID	Name	Sex	Age	Height	Weight	Team	region	Games	Year	Season	City	Sport
		Michelet											
194755	68922	Leroy	M	<NA>	<NA>	NaN	Mascaret-4	France	1900 Summer	1900	Summer	Paris	Sailing
194759	75648	Charles William Martin	M	71	<NA>	NaN	Pirouette-5	France	1900 Summer	1900	Summer	Paris	Sailing

1385 rows × 15 columns

In [47]:
```
df.isnull().sum()
```

Out[47]:
```
ID            0
Name          0
Sex           0
Age        9474
Height    60171
Weight    62875
Team          0
region      349
Games         0
Year          0
Season        0
City          0
Sport         0
Event         0
Medal         0
dtype: int64
```

Clean the data

In [48]:
```
df.dropna(inplace=True)
df.drop_duplicates(inplace=True)
df.head()
```

Out[48]:

	ID	Name	Sex	Age	Height	Weight	Team	region	Games	Year	Season	City	Sport	Event	Medal
38	133735	Karol Marian Zajc (-Sayons)	M	22	175	74.0	Poland	Poland	1936 Winter	1936	Winter	Garmisch-Partenkirchen	Alpine Skiing	Alpine Skiing Men's Combined	NA
58	2792	mile Allais	M	23	168	68.0	France	France	1936 Winter	1936	Winter	Garmisch-Partenkirchen	Alpine Skiing	Alpine Skiing Men's Combined	Bronze
83	3022	Pentti Johannes Alonen	M	22	172	65.0	Finland	Finland	1948 Winter	1948	Winter	Sankt Moritz	Alpine Skiing	Alpine Skiing Men's Combined	NA
92	23517	James Couttet	M	26	168	67.0	France	France	1948 Winter	1948	Winter	Sankt Moritz	Alpine Skiing	Alpine Skiing Men's Combined	Bronze
101	12204	Robert "Bob" Blatt, Jr.	M	26	181	76.0	United States	USA	1948 Winter	1948	Winter	Sankt Moritz	Alpine Skiing	Alpine Skiing Men's Combined	NA

Now Analyse the basics facts

In [86]:
```
import matplotlib as plt
import matplotlib.pyplot as plt
```

1. Total winning by male and female

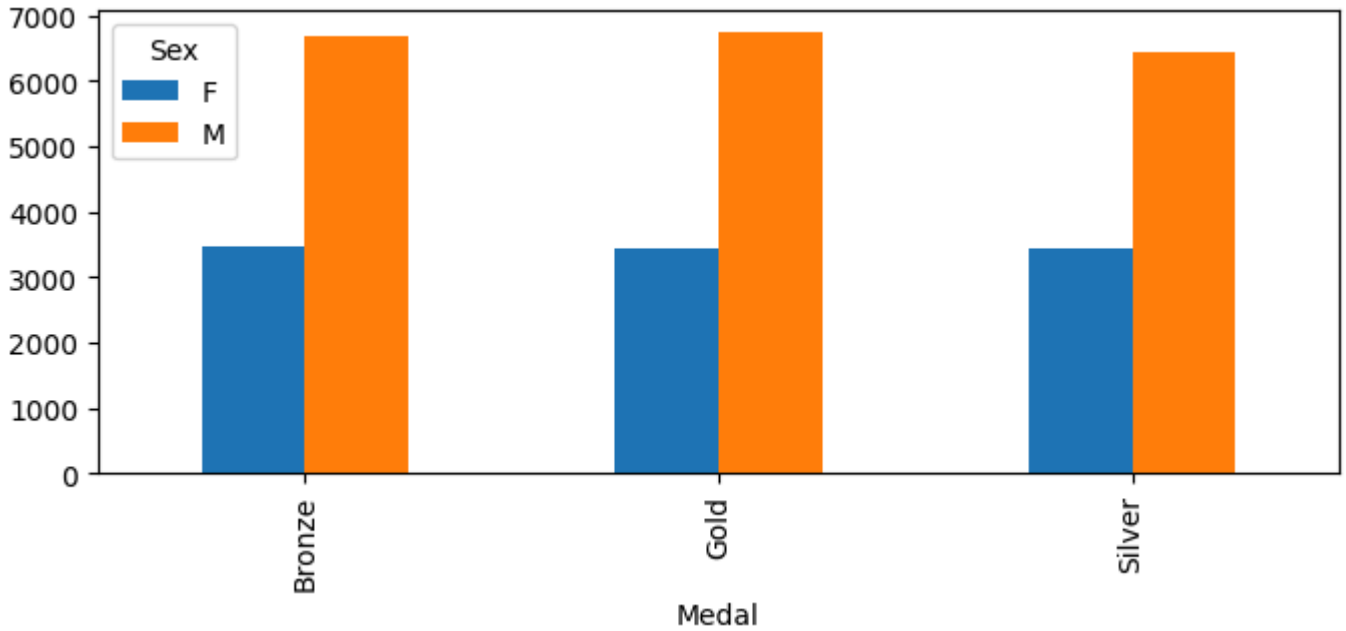
In [94]:
```
table1 = df[df["Medal"] != 'NA'].groupby(["Medal","Sex"]).size().unstack()
table1
```

Out[94]:

Sex	F	M
Medal		
Bronze	3471	6673
Gold	3437	6729
Silver	3435	6427

In [96]:
```
table1.plot(kind="bar", figsize=(8,3))
```

Out[96]: <Axes: xlabel='Medal'>



2. pivot table to check the history of winning trends by male and female over the year

In [85]:
```
table2 = df[df["Medal"] != 'NA'].groupby(["Year","Sex"]).size().unstack()
table2
```

Out[85]:

Sex	F	M
Year		
1896	NaN	20.0
1900	NaN	38.0
1904	NaN	59.0
1906	NaN	69.0
1908	NaN	134.0
1912	NaN	130.0
1920	1.0	181.0
1924	4.0	181.0
1928	20.0	150.0
1932	25.0	151.0
1936	23.0	170.0
1948	24.0	171.0
1952	52.0	250.0
1956	103.0	487.0
1960	175.0	800.0
1964	236.0	940.0
1968	259.0	989.0
1972	293.0	1108.0
1976	435.0	1063.0
1980	477.0	1095.0
1984	545.0	1138.0
1988	614.0	1213.0

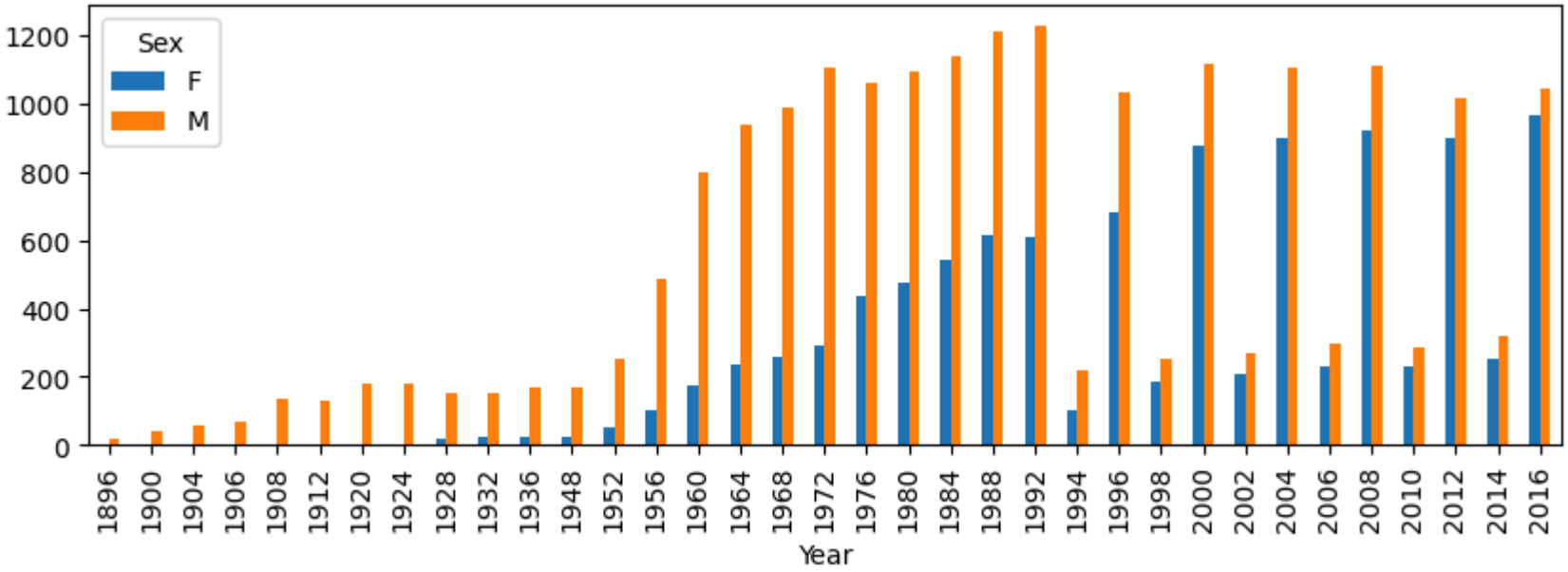
Sex	F	M
Year		
1992	607.0	1227.0
1994	104.0	220.0
1996	684.0	1033.0
1998	187.0	250.0
2000	877.0	1116.0
2002	207.0	270.0
2004	897.0	1103.0
2006	231.0	295.0
2008	921.0	1111.0
2010	228.0	287.0
2012	897.0	1014.0
2014	250.0	320.0
2016	967.0	1046.0

In [102…

```
table2.plot(kind="bar", figsize=(10,3))
```

Out[102…

```
<Axes: xlabel='Year'>
```



3. pivot table to check the history of winning diffrent medal trends by male and female over the year

In [104…

```
table3 = df[df["Medal"] != 'NA'].groupby(["Year", "Medal", "Sex"]).size().unstack()
table3
```

Out[104…

Year	Sex / Medal	F	M
1896	Bronze	NaN	3.0
	Gold	NaN	11.0
	Silver	NaN	6.0
1900	Bronze	NaN	9.0
	Gold	NaN	19.0
…	…	…	…
2014	Gold	85.0	105.0
	Silver	82.0	108.0
2016	Bronze	330.0	370.0
	Gold	318.0	343.0
	Silver	319.0	333.0

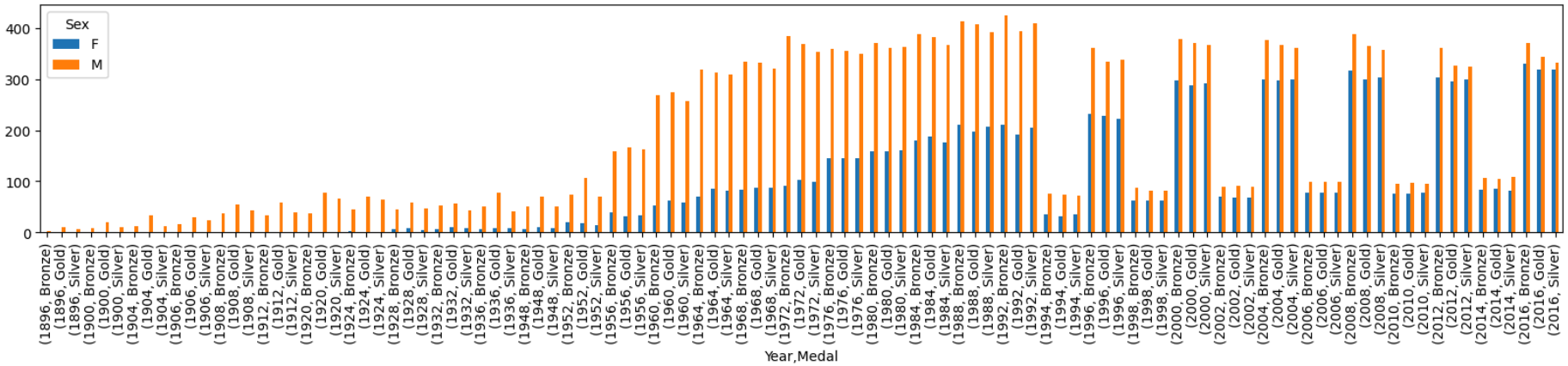
105 rows × 2 columns

In [106…

```
table3.plot(kind="bar", figsize=(20,3))
```

Out[106…

```
<Axes: xlabel='Year,Medal'>
```


Sex
F
M
400
300
200
100
0
(1896, Bronze)
(1896, Gold)
(1896, Silver)
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(1900, Silver)
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(2016, Bronze)
(2016, Gold)
(2016, Silver)
Year,Medal